


$$\begin{array}{lcl}
 & & p_i \\
 \downarrow & i = l & (1) \quad p_l \\
 & i = m & (2) \quad p_m \\
 & i = n & (3) \quad p_n
 \end{array}$$

$$P_i = \sum_{i=1}^n p_i$$

$$\sum_{i=1}^1 p_i = p_l$$

$$p_l + p_m$$

$$p_l + p_m + p_n = 1$$

$$H_i = \frac{1 - P_i}{p_i}$$

$$\frac{1 - p_l}{p_l} = \frac{p_m + p_n}{p_l}$$

$$\frac{1 - p_l - p_m}{p_m} = \frac{p_n}{p_m}$$

$$\frac{1 - 1}{p_n} = 0$$

A is sufficient for B : $A \Rightarrow B$

A is necessary for B : $B \Rightarrow A$

A is necessary & sufficient for B : $A \Leftrightarrow B$

to be read as "equivalent"
OR if and only if

$$\max_{(q_i, t_i)_{i \in \{l, m, n\}}} \sum_{i \in \{l, m, n\}} p_i (\ln(q_i) - t_i) \rightarrow p_e (\ln(q_e) - t_e) + (p_m + p_n) (\ln(\bar{q}) - \bar{t})$$

$$\text{s.t.} \quad t_n - c_n \bar{q} = 0$$

$$t_e - c_e q_e = t_m - c_e \bar{q}$$

$$t_m - c_m \bar{q} = t_n - c_m \bar{q}$$

$$q_e > \bar{q} = q_m = q_n$$

$$q_i \geq 0 \quad i = l, m, n$$

$$q_e > q_m, q_n \Rightarrow q_m \geq q_n \text{ is binding}$$

$$\Leftarrow q_m = q_n = \bar{q}$$